

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. We construct the compatible hard-sphere Green trace graph using the normal-flux measure exactly once and analyze recollisions on the full fixed horizon. For every fixed cyclic genealogy the actual linearized collision law has an explicitly witnessed transverse analytic minor; genealogy-dependent sublevel exponents are then summed by dominated convergence under one factorial trajectory majorant. The limiting marked Hamiltonian is derived directly from the global tree series. Smooth positive balanced pairs are exposed by actual contact sources, a continuous kinetic Hodge repair supplies rate-dense recovery, and full grand-canonical and source-conditioned microcanonical density–contact LDPs follow.

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1. ROUND-FOURTEEN POSITIVE CLOSURE: COMPATIBLE TRACES, TRUE JACOBI TUBES, AND A FIXED-HORIZON JOINT LDP

The grand-canonical construction in this section is logically prior to B1. The whole interval $[0, T]$ is treated in one time-ordered trajectory expansion; no per-block error is sewn. A cyclic genealogy is removed only after its actual linearized collision map has been analyzed. Constants and sublevel exponents may depend on the fixed genealogy, and the absolutely summable tree majorant permits dominated convergence over all genealogies.

1.1. The compatible kinetic trace graph. For $k \geq 1$, let

$$\Omega_k^\varepsilon = \{Z_k : |x_i - x_j| > \varepsilon, i \neq j\}$$

and let Σ_{ij}^- be the incoming part of the contact face $|x_i - x_j| = \varepsilon$. A normal-flux trace is already the measure

$$d\gamma_{ij}^- = ((v_i - v_j) \cdot \omega)_+ f_k^- dt d\sigma_{ij} dV_k.$$

Its norm is

$$\|\gamma_k^-\|_{\text{tr}} = \sum_{i < j} \int w_k d|\gamma_{ij}^-|,$$

with no second multiplication by the normal velocity.

Theorem 1.1 (Closed Green trace graph). *Let $\mathfrak{X}_{\alpha,\beta}$ be the factorially weighted hierarchy space and let $\mathfrak{D}_{\alpha,\beta}$ consist of hierarchy densities whose transport divergence and incoming normal-flux traces have the same weighted finite mass. Then $\mathfrak{D}_{\alpha,\beta}$ is closed for the graph norm, and*

$$\begin{aligned} & \int_{\Omega_k^\varepsilon} (\partial_t + V_k \cdot \nabla_{X_k}) \phi df_k \\ &= \langle f_k(T), \phi_T \rangle - \langle f_k(0), \phi_0 \rangle - \sum_{i < j} \int_{\Sigma_{ij}^-} (\phi(Z'_k) - \phi(Z_k)) d\gamma_{ij}^-. \end{aligned}$$

Weak graph convergence preserves all simple-contact traces. Multiple-contact corners are retained as lower-dimensional graph strata and have zero simple-flux mass.

Proof. Apply the Gauss–Green theorem to smooth truncations of Ω_k^ε avoiding pairwise intersections of contact faces. The outward transport flux is $(V_k \cdot n) f_k d\sigma dV$; pairing incoming and outgoing faces by the elastic reflection gives the displayed jump term with the single flux measure γ^- . Uniform graph-norm bounds give weak compactness of the interior currents and their normal traces in the divergence-measure topology. Passing through the truncations proves closedness. The intersection of two distinct contact faces has codimension two and carries no mass for a finite normal trace; it remains recorded for iterated traces but contributes no extra factor to the simple Green formula. $\square$

1.2. Real trajectory genealogies. A connected trajectory genealogy $\mathcal{G}$ records labels, ordered creation contacts, and surplus contacts. On a regular compact piece $\mathcal{K}_{\mathcal{G}}$, collision times are strictly ordered, all incidence cosines exceed a positive genealogy-dependent number, and velocities are bounded. The coarea variables are initial roots, creation times, incoming relative velocities, and collision normals.

Let t_* be the first surplus contact. Remove only its equation, not its reflection or future trajectory. The resulting pseudo-orbit agrees with the true orbit up to t_* . Let

$$\Psi_{\mathcal{G}}(\xi) = \Pi_{\omega^\perp_*}(x_a(t_*^-; \xi) - x_b(t_*^-; \xi))$$

be its two transverse contact coordinates.

Lemma 1.2 (Fixed-genealogy physical Jacobi transversality). *For every fixed cyclic genealogy $\mathcal{G}$, the regular parameter domain admits a finite cover by analytic charts and a finite exceptional analytic set such that on each nonexceptional chart one 2×2 minor of $D\Psi_{\mathcal{G}}$ is not identically zero. Consequently there are $C_{\mathcal{G}} < \infty$ and $\beta_{\mathcal{G}} > 0$ with*

$$\mathfrak{m}_{\mathcal{G}}\{s_{\min}(D\Psi_{\mathcal{G}}) < \rho\} \leq C_{\mathcal{G}}\rho^{\beta_{\mathcal{G}}}.$$

No exponent uniform in genealogy depth is asserted.

Proof. Differentiate the actual collision law. At a collision with incoming gap $g = (v - v_*) \cdot \omega > 0$,

$$Dv^+ = Dv^- - ((Dv^- - Dv_*^-) \cdot \omega + (v - v_*) \cdot D\omega)\omega - g D\omega,$$

and analogously for v_*^+ . Thus the two tangential components of $D\omega$ enter the outgoing relative velocity with leading map $-2gI_{\omega^\perp}$.

Choose the youngest creation collision whose two descendant branches contain the labels meeting at t_* . The collision normal there is one of the coarea variables. To show that a transverse minor is not identically zero, construct one witness configuration for the fixed ordered graph: put all subsequent creation contacts in a short interval of length δ , choose their normals and velocities generically, and leave a final free-flight interval of length $L > 0$ before t_* . The derivative of the surplus transverse position with respect to the two tangential components of the chosen normal is

$$-2gL I_{\omega^\perp} + O(\delta).$$

For δ small this minor is nonzero. Earlier and later collision derivatives are the full physical Jacobi matrices displayed above; they have not been replaced by fixed-normal reflections or by symplectic rank alone.

The entries of the minor are real analytic on every regular chart. Since the determinant is not identically zero, the quantitative analytic sublevel theorem on the compact chart gives an exponent $\beta_{\mathcal{G}} > 0$. A finite chart cover proves the estimate. $\square$

Lemma 1.3 (Surplus contact tube for a fixed genealogy). *The absolute activity of a fixed cyclic genealogy satisfies*

$$|\zeta_\varepsilon(\mathcal{G})| \leq M_{\mathcal{G}} \varepsilon^{\alpha_{\mathcal{G}}}, \quad \alpha_{\mathcal{G}} > 0,$$

after velocity and grazing truncation. The estimate includes every bounded particle and actual-contact source derivative.

Proof. On the set where $s_{\min}(D\Psi_{\mathcal{G}}) \geq \rho$, coarea bounds the two-dimensional contact tube by $C\varepsilon^2\rho^{-2}$. On its complement use Lemma 1.2. Optimizing $\rho^{\beta_{\mathcal{G}}} + \varepsilon^2\rho^{-2}$ gives $\alpha_{\mathcal{G}} = 2\beta_{\mathcal{G}}/(\beta_{\mathcal{G}} + 2) > 0$. Incoming flux controls local grazing denominators, and the regular truncations are removed by monotone convergence. Bounded source insertions multiply the fixed-genealogy majorant by a finite factor. $\square$

1.3. The global fixed-horizon sum. Let $\mathfrak{T}$ be the set of creation trees and $\mathfrak{C}$ the cyclic genealogies. Collision coarea and the ordered time simplex give

$$M_{\mathcal{G}} \leq \frac{k! C^k T^{k-1+c(\mathcal{G})}}{(k-1+c(\mathcal{G}))!} e^{C \sum_i |v_i|^2},$$

and the sum of these majorants over all labelled connected genealogies is finite for $T \leq T_0$.

Theorem 1.4 (Fixed-horizon all-contact cluster theorem). *For $T \leq T_0$, the grand-canonical log-Laplace functional with bounded particle-path and actual-contact sources has an absolutely normally convergent connected trajectory expansion on a common complex source ball. Creation trees converge to the Boltzmann collision-tree series, and*

$$\sum_{\mathcal{G} \in \mathfrak{C}} |\zeta_\varepsilon(\mathcal{G})| \longrightarrow 0.$$

The convergence is locally uniform with every fixed source derivative.

Proof. The creation-tree bound follows by iterating collision coarea and integrating the ordered time simplex. It is a summable majorant independent of ε . For every fixed cyclic genealogy, Lemma 1.3 gives convergence to zero. Dominated convergence over the countable genealogy set therefore removes the entire cyclic sector, without a genealogy-uniform Łojasiewicz exponent and without a mesh factor. Tree terms converge by dominated collision coarea. Normal convergence permits all fixed source derivatives. $\square$

In particular the limiting cumulant is constructed directly by the global tree series. It is not defined through the B4 semigroup.

Theorem 1.5 (Constructed marked Hamiltonian). *The limiting pressure is the unique analytic mild solution on the short-time chart with Hamiltonian*

$$\mathbb{H}(f, p, \psi) = \langle f, v \cdot \nabla_x p \rangle + \frac{1}{2} \int f f_* ((v - v_*) \cdot \omega)_+ (e^{\Delta_{\omega} p + \psi} - 1).$$

Every ψ -derivative is a factorial moment of the actual incoming contact measure.

Proof. In a limiting tree single out its last creation contact. Deleting it leaves two independent rooted subtrees; collision coarea gives the incoming flux and source multiplication gives $e^{\Delta p + \psi} - 1$. This derives the mild equation term by term. The cluster norm makes Picard iteration contractive for $T \leq T_0$. Because ψ marked the microscopic normal-flux trace, its derivative selects actual contacts. $\square$

1.4. Exposure and lower recovery. For a smooth positive balanced pair put

$$dA_f = \frac{1}{2} f f_* ((v - v_*) \cdot \omega)_+ dt dx dv dv_* d\omega, \quad q = \frac{d\Gamma}{dA_f}.$$

Lemma 1.6 (Regular exposed pair). *If q and q^{-1} are bounded with Gaussian velocity weights and (f, Γ) satisfies the exact weak balance, there are bounded smooth sources p, ψ with*

$$\psi = \log q - \Delta p$$

such that the first derivative of the pressure in Theorem 1.5 is exactly (f, Γ) . Under the corresponding microscopic tilt, the empirical density/contact pair concentrates exponentially at (f, Γ) .

Proof. Given q , solve the biased Boltzmann equation forward from the chosen positive initial datum. On the short-time Maxwellian chart its collision operator is locally Lipschitz in the Gaussian weighted norm. Solve the backward adjoint equation for p with the desired terminal density source, and define ψ by the displayed identity. Differentiating the tree mild equation gives the forward biased equation and hence the prescribed pair. Strict convexity of $e^z - 1$, together with the positive initial covariance, makes the Hessian positive on the balanced quotient. The normal complex cluster expansion and Chernoff bounds then give exponential concentration. $\square$

Lemma 1.7 (Continuous kinetic Hodge repair). *Every finite-action balanced pair is approximated, with convergence of entropy action and endpoints, by smooth positive exposed pairs.*

Proof. First truncate velocities and convolve in time and space away from endpoints. Let b be the resulting small balance defect. On each local energy-momentum fiber use the collision Dirichlet form

$$\mathcal{E}_f(r, \phi) = \int \Delta r \Delta \phi dA_f.$$

On the orthogonal complement of the five collision invariants, the Maxwellian lower bound gives coercivity. The compatibility of b with mass, momentum, and energy follows from the original balance. Lax–Milgram therefore gives r with $\mathcal{E}_f(r, \phi) = b(\phi)$. Set $\delta\Gamma = (\Delta r)A_f$. It has the exchange/pre-post symmetry and repairs the balance exactly. Add a Maxwellian background before solving and let its size decrease after the smoothing scale; then $q + \Delta r > 0$, and the entropy change tends to zero by Orlicz equi-integrability. Endpoint correctors are transported in short boundary layers. $\square$

1.5. The full grand-canonical and microcanonical LDPs. Define

$$\ell(q) = q \log q - q + 1.$$

Theorem 1.8 (Grand-canonical joint LDP). *The density and actual-contact pair satisfies a good LDP at speed μ_ε with rate*

$$I_T^{\text{gc}}(f, \Gamma) = I_0^{\text{gc}}(f_0) + \begin{cases} \int \ell(d\Gamma/dA_f) dA_f, & (f, \Gamma) \text{ balanced and } \Gamma \ll A_f, \\ +\infty, & \text{otherwise.} \end{cases}$$

Proof. Bounded-source upper bounds follow from Theorem 1.4. Increasing positive particle and contact tests give exponential compact containment. Scaling the balance test makes the dual infinite off the closed trace graph. Monotone truncation of $\log q$ exhausts the full entropy conjugate.

For the lower bound, use Lemma 1.6 on smooth positive pairs and compute the exact Radon–Nikodym cost, which is the displayed entropy. Lemma 1.7 supplies rate-dense exposed recovery. The standard compact-cover argument completes open-set lower bounds and goodness. $\square$

Theorem 1.9 (Microcanonical transfer and Round-fourteen B2 closure). *After the source-dependent exact-number and continuous-shell conditioning of B1, the same pair satisfies a good LDP with*

$$I_T^{\text{mc,a}}(f, \Gamma) = I_{0,a}(f_0) + \int \ell(d\Gamma/dA_f) dA_f$$

on the balanced domain. The dependency order is

$$B2\text{--}GC \longrightarrow B1 \longrightarrow B2\text{--}MC.$$

Proof. B1 applies the exact shell coefficient to the full joint source pressure proved in Theorem 1.4. Its source-dependent change of measure transfers both the upper bound and every exposed lower recovery. Only the initial rate is replaced by its constrained information projection; the dynamic entropy is unchanged. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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